Alert Governance Stress-Test

Technical report, ITU AI for Good Lab Hackathon (UAE). Team: [TEAM NAME / MEMBERS]. Code and run records: [PUBLIC REPOSITORY URL].

Scope statement. This project does not assess UAE institutions. It tests whether a published-policy knowledge base enables accountable, evidence-grounded coordination during a hypothetical public-alert governance failure. The scenario is fictional (district “Northvale”).

(a) Introduction to the agentic solution

Institutions are modelled as AI agents (Telecom/Operator, Emergency Management, Policy Adviser) that receive private, role-specific event notices from a six-event fictional false-public-alert scenario. Each agent retrieves from a local knowledge base (KB) of 584 chunks built only from officially published documents, and returns structured records: claims, evidence labels, flags, and a recommended action. Four controls make the output accountable: (1) an **evidence gate** rejects any claim without a cited KB passage; (2) a **claim-level verifier** (a second, cheaper model) checks that each cited passage supports the claim; (3) an **append-only, hash-chained audit log** records every record; (4) a **human approval gate**: agents only propose, a named human approves or rejects. A gap analyst then reads the flagged records and writes an evidence-linked gap report. A plain LLM with the same inbox but no KB runs as a baseline and is scored by the same verifier. The system does not decide who is right. It shows what the published evidence supports, what it does not establish, and where human authority is required.

(b) Use case description

A cell-broadcast “Emergency warning” reaches about 40,000 handsets; the operator log shows no matching request and no authenticated originator (E1, E3). Residents and journalists ask whether it is real (E2, E4, E6); a gateway vendor reports a test-mode setting left enabled and that it cannot recall delivered messages (E5). The question for the agents is who can validate, correct or cancel the alert, on what published basis, and what the sources do not establish. The scenario is built to stress governance, not technology: every event is plausible but invented, and no real internal information is used.

(c) Use case requirements

- **Evidence over assertion:** no claim about a mandate without a cited, in-force, verifiable passage; absence of evidence is reported as “not established from public sources”, never filled in.
- **Authority-aware retrieval:** Arabic text of the NCEMA law is authoritative; its English text is carried only as a labelled unofficial translation, capping claims on it at partially_supported. A coordination clause is never used as proof that a body owns, triggers, authorises or cancels an alert. The repealed 2024 law appears only as labelled history.
- **Accountability:** tamper-evident audit trail, named human approval, no autonomous action. **Reproducibility:** pinned model snapshots, seeded sampling, disk-cached calls. **Locality:** local embeddings (bge-small-en-v1.5, ONNX); only the model calls leave the machine. **No blame wording** about real institutions.

(d) Mapping to ITU-T Y.3172 (06/2019)

Y.3172 specifies an architectural framework for machine learning in future networks (IMT-2020). We use its node vocabulary as a structural analogy for our pipeline; we do not claim conformance, since our system is not a network function and has no ML underlay network.

Y.3172 node (clause 8.1)	Y.3172 role	Counterpart in this system
SRC (source)	Source of data for the pipeline	Scenario events and each agent’s private inbox
C (collector)	Collects data from SRC nodes	Runtime assembling inbox + anchor retrieval from the KB
PP (preprocessor)	Cleans or aggregates data for the model	Chunking, authority metadata, mode and jurisdiction filters
M (model)	The ML model	Agent LLM and the cheaper verifier LLM
P (policy)	Applies policies to model output	Evidence gate, claim verifier, action gate, flag validation
D (distributor) / SINK	Distributes output to the target that acts	Audit log and human approval queue; the human, not the system, acts
MLFO	Orchestrates and chains pipeline nodes	Scenario engine: event order, inbox routing, one bounded follow-up round
ML sandbox	Isolated domain to test pipelines before live use	The whole exercise runs on a fictional scenario; nothing touches a live system

(e) Mapping to ITU AI Readiness Report 2.0

The report's basic framework has five dimensions: Data, Digital Infrastructure, Digital Skills, Innovation Ecosystem and AI Policy; AI Policy covers AI Policy and Regulation, Regulatory Quality and Implementation. Our work touches two directly and one indirectly:

- **AI Policy / Implementation and Regulatory Quality:** the gap report is an evidence-linked check of whether published rules identify a decision owner and a correction path; it is a test method, not a readiness score.
- **Data / Data Governance (accountability safeguards):** every claim carries source, article, translation status and verifier status; audit records are append-only and hash-chained.
- **Innovation Ecosystem / Standards:** the report lists AI pipeline standards (Y.3172) as a metric; our mapping in (d) is an example of using it. We make no readiness claim about any country.

(f) Requirements mapped to published local policies

Only articles we read in the source documents are listed. Items marked * rest on the unofficial English translation of the NCEMA law; the Arabic original prevails and should be checked.

Requirement	Source and provision	How the agent enforces it
Public warning system competence, in coordination with NCEMA	Federal Decree-Law 8/2025, Art. 4(5) (official English translation; Arabic prevails; in force 1 Jan 2026)	Cited as the in-force competence; wording kept to what the article says
Operator must comply with directions of TDRA or another competent authority in a declared public emergency	TDRA Public Telecommunication Licence 3/2026, Art. 8	Telecom agent cites it; it may not name which authority is "competent" (flagged instead)
Warning service uses cell broadcast; alerts recognisable, Arabic and English	TDRA TS FS 001 v2.0 (UAE-Alert), Sections 2, 3, 4.2	Technical context only; never used as mandate evidence
Coordination, liaison and unified media message*	NCEMA Decree-Law 2/2011, Arts. 13, 14, 23 to 26	Gate treats these as coordination clauses; claims capped at partially_supported
Repealed civil-defence law	Decree-Law 35/2024, repealed by 8/2025 Art. 19(1)	Retrievable only in history mode, labelled as repealed
Global comparison (not UAE rules)	EU 2018/1972 Art. 110; FCC Hawaii 2018 report; ITU-T X.1303 bis (CAP)	Must cite a reference_global chunk; never presented as UAE obligations

(g) Evaluation scenario and results

Six events (E1 to E6), three roles, private inboxes and one bounded follow-up round. The baseline gets the same inbox and the same jurisdiction line but no KB. Both are scored by the same verifier model against the same KB. **One run per condition; counts, not statistics; no significance is claimed.**

Metric (run run-20261108-200619-f7608f8d)	Plain LLM	KB-grounded
Claims made	28	151
Citation validity, strict (uncited claim = not valid)	9/28 (32.1%)	146/151 (96.7%)
Validity among claims naming a specific provision	9/16 (56.2%)	146/151 (96.7%)
Mandate claims passing verification	4/16	86/91
Unsupported mandate claims left in output	12	0
Topics the sources do not establish, correctly flagged (of 4)	4	4
Gaps linked to chunk-level evidence	0 (none possible)	5 (3 gaps, 2 lower-confidence)

Reading the result. Both conditions flagged all four "not established" topics, so the KB's advantage is not abstention; it is that claims carry checkable citations. 12 of the baseline's 28 claims named no provision at all (honest admissions, but unsupported when asserted as a mandate). Retrieval was measured on 17 questions: hit@1 12/17, hit@3 14/17, hit@5 16/17; one miss (Q06) and two lower-ranked hits (Q07, Q12) are reported as they are. **Limits:** the same model family writes and verifies, so a human spot check of 20 claims is the independent check: [SPOT-CHECK RESULT: TO BE FILLED]. We changed three evaluation settings after seeing early results (baseline jurisdiction and token budget, abstention buckets, an absence test for gaps); all are recorded in docs/CHANGELOG_DECISIONS.md.

(h) Policy and standards gaps identified

Each gap below was produced by the run, traced to flagged audit records, and passed a check that the cited passages are on topic and do not establish the point, plus a 12-result search of current UAE sources that found nothing that does. They describe what the public documents establish, not how any institution behaves. Each depends on a corpus without the NCEMA Executive Regulation, which the law refers to for event classification and body roles.

ID	Gap (published sources do not establish…)	Key evidence and caveat
G1 (high)	an event-specific decision owner for originating, authorising, validating or directing a public warning; texts say “authorities”, “TDRA or other competent authority”, “relevant bodies according to competence”	TS FS 001 pp. 3-4; Licence Art. 8; NCEMA Arts. 13, 14*. Evidence label: partially_supported
G2 (high)	a public procedure for cancelling, correcting or publicly repudiating an erroneous alert after delivery	TS FS 001 p. 4 (a user can dismiss a message; no authority-led cancellation); Licence Art. 8; NCEMA Arts. 14, 24, 25*. Comparison: FCC Hawaii report on cancellation not recalling delivered messages. partially_supported
G4 (medium)	a verification and escalation channel for municipal or public-facing offices asked about an unconfirmed alert	NCEMA Arts. 23 to 26* provide general liaison and media coordination only. partially_supported

Two further items are reported as **lower-confidence observations**, not firm gaps: the published framework does not describe authenticated-originator or gateway audit requirements for operators (G3), and does not describe safeguards that keep internal test messages from reaching live broadcast (G5). The sources never promise such detail, so their absence is a question for the documents’ owners, not a defect we can show. Detail left to the Executive Regulation, and the fact that the NCEMA law is published in Arabic with only an unofficial English translation, are listed as **documentation limitations**, not gaps.

Where human authority is required: naming the competent authority, approving any correction, and checking every NCEMA citation against the Arabic text. The full gap report, scorecard, spot-check sheet, audit records and a local timeline viewer are in the repository.